

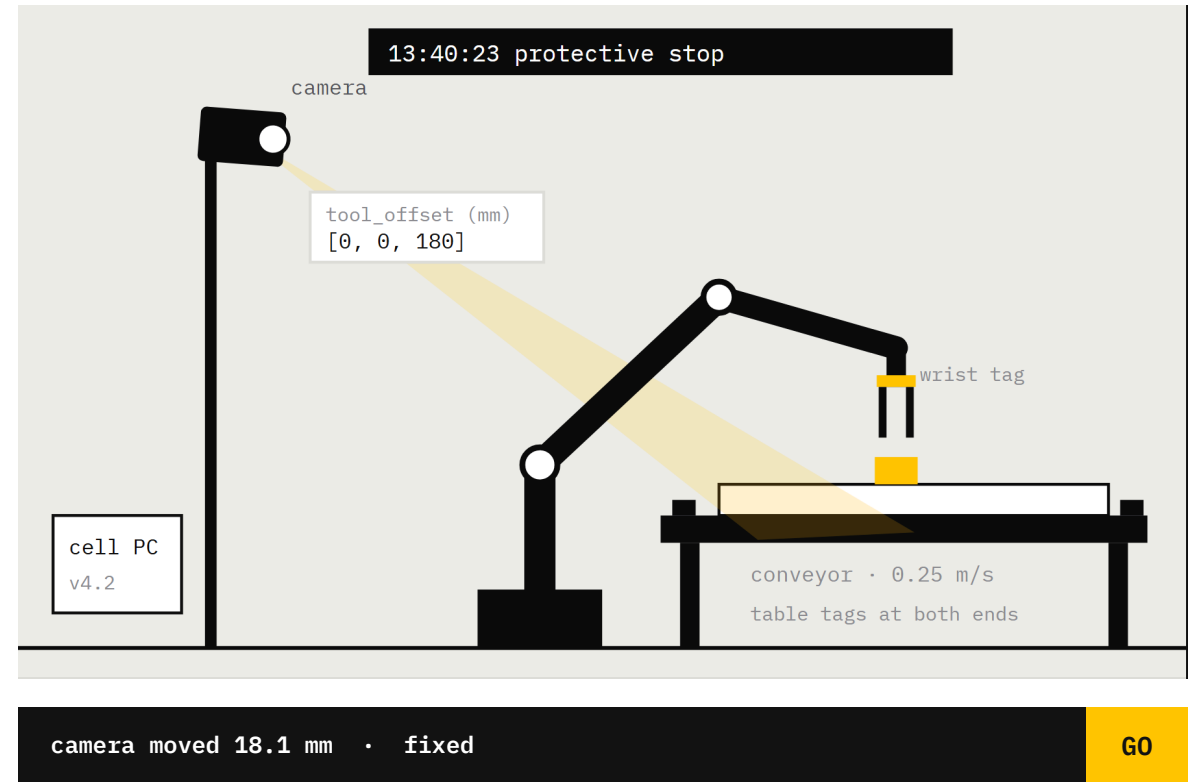
Kintrace

Kintrace finds what moved on a robot, fixes it, and tells you it is good to go.

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Previously LoopCell. Same product, new name.



The tool's own output, from a run on our simulator.

Kintrace finds what moved on a robot, fixes it, and tells you it is good to go.

13:40:23

something changed

Notices

It sees the exact moment a robot got knocked out of place, from data the robot already records.

18.1 mm

the camera moved

Explains

It says what moved and by how much. Not an alert. A measurement.

6 / 6

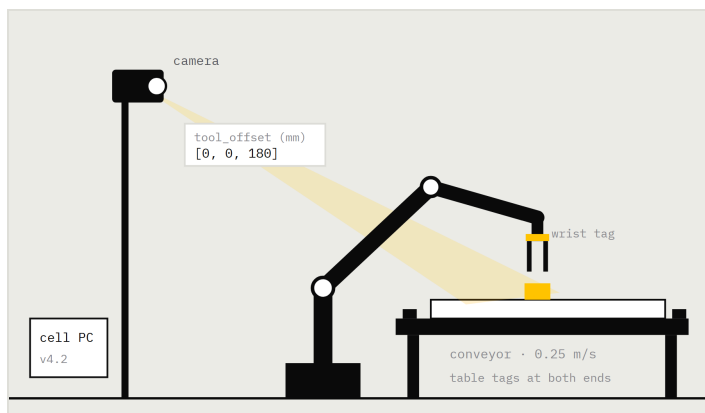
test grabs, then GO

Fixes

It updates the robot, runs test grabs, and says whether the cell is good to go.

Software only. No new hardware. It works on robot arms that already have a camera.

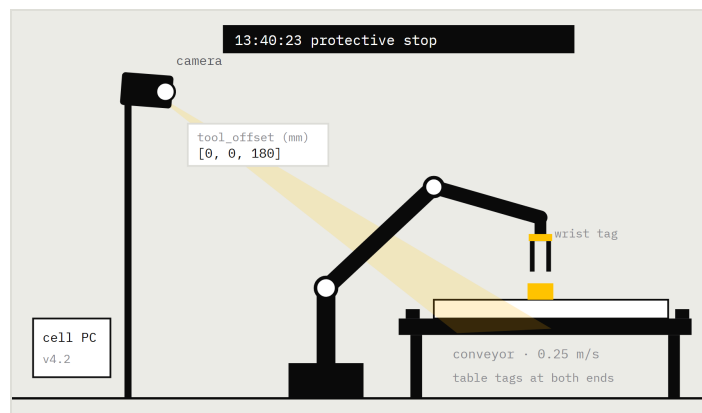
At 1:40 a cart clips a camera. Nobody finds out until 2:10.



1:40

Running fine

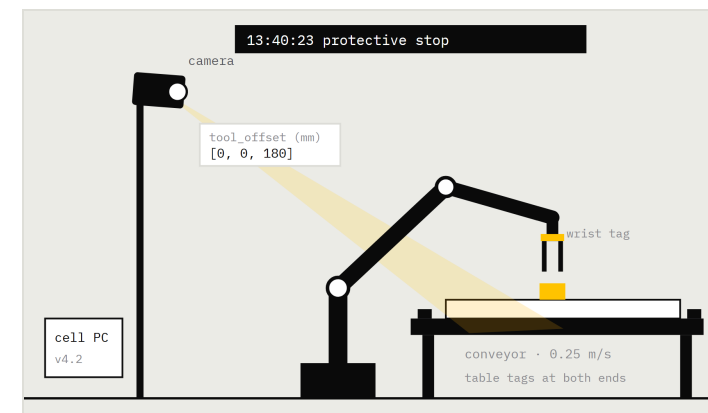
The camera is where the calibration says it is.



1:40 to 2:10

Bumped, and nobody knows

Every grab is a centimetre off. The robot still reports healthy.

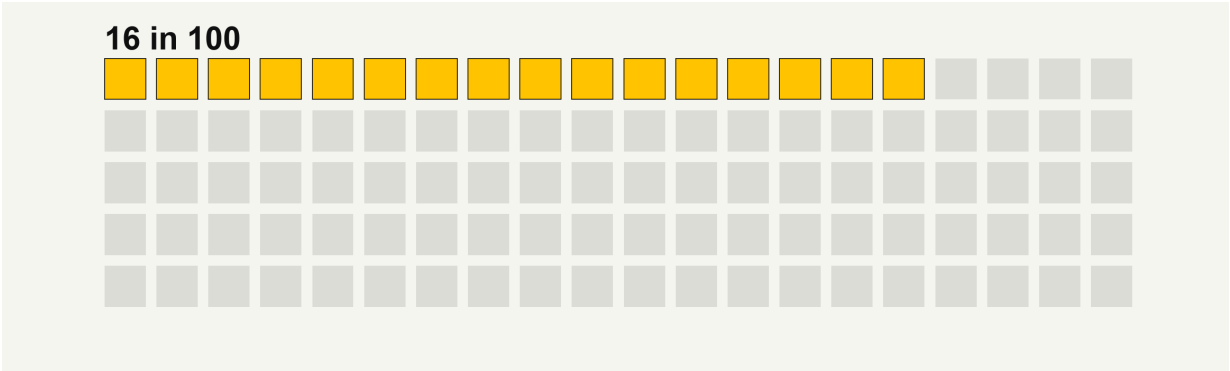


1:43 with Kintrace

Found, fixed, signed off

Camera moved 18.1 mm. New calibration written. Six test grabs.

You pay for it twice: in ruined training data, and in stopped lines.



Robot training data

An open audit of 4,959 recorded episodes flagged 15.6% of them for exclusion, and another half for review.

\$2.3M

an hour

Up to, for an idle production line at a big car plant. Across the 500 largest manufacturers, unplanned downtime of every kind costs \$1.4 trillion a year, about 11% of revenue.

Everything on the market tells you something is wrong, not what moved.

Monitoring dashboards

Ember, Foxglove, Formant

They watch the software and the logs. They will tell you a cell is slow or a run failed. They do not measure where the camera is, so they cannot name the physical cause.

Calibration hardware

Dynalog, CAPTRON, Renishaw

These do measure, and they are accurate. But you bolt a device into every cell you want covered, and each one mostly does the tool.

A technician

How it is actually done

Someone drives out and rules out each cause by hand, one at a time, with the cell down while they do it. It works. It costs hours.

We have not found anyone doing all of it from software alone: when it changed, what moved, by how much, and the fix written back.

What each of these does is taken from their own sites and materials. Listed in [pitch/SOURCES.md](#).

This is the tool's own output, not a mockup.

\$ KINTRACE INCIDENTRECORD.TXT

KINTRACE recommission record, Cell 3

=====

Changed at : 13:40:23 (lines up with protective stop at 13:40:23: collision detected, arm stopped)

seen in : camera vs arm at 13:40:23

seen in : camera vs table marker at 13:40:23

seen in : first missed pick at 13:40:25

What changed (6 s check motion)

- Camera moved: 18.1 mm, 0.04 deg (parts appear 18 mm off in the pick area)

Fix written to controller

- camera calibration updated

Verification (check motion re-run)

[ok] Command timing 8 ms vs 8 ms normal

[ok] Camera vs fixed cell marker table marker where it should be (1.0x noise)

[ok] Camera vs joint encoders disagreement 1.0x sensor noise

[ok] Fingertip vs known-good fingertip 0.1 mm from its known-good position

[ok] Tool offset setting matches known-good

test picks: 6/6 ok

STATUS: GO

GO

CELL BACK IN PRODUCTION

What we actually did: bump the camera, sized for about 20 mm at the fingertip.

- 1

Notices

Finds the moment things changed and lines it up with the protective stop the controller logged.
- 2

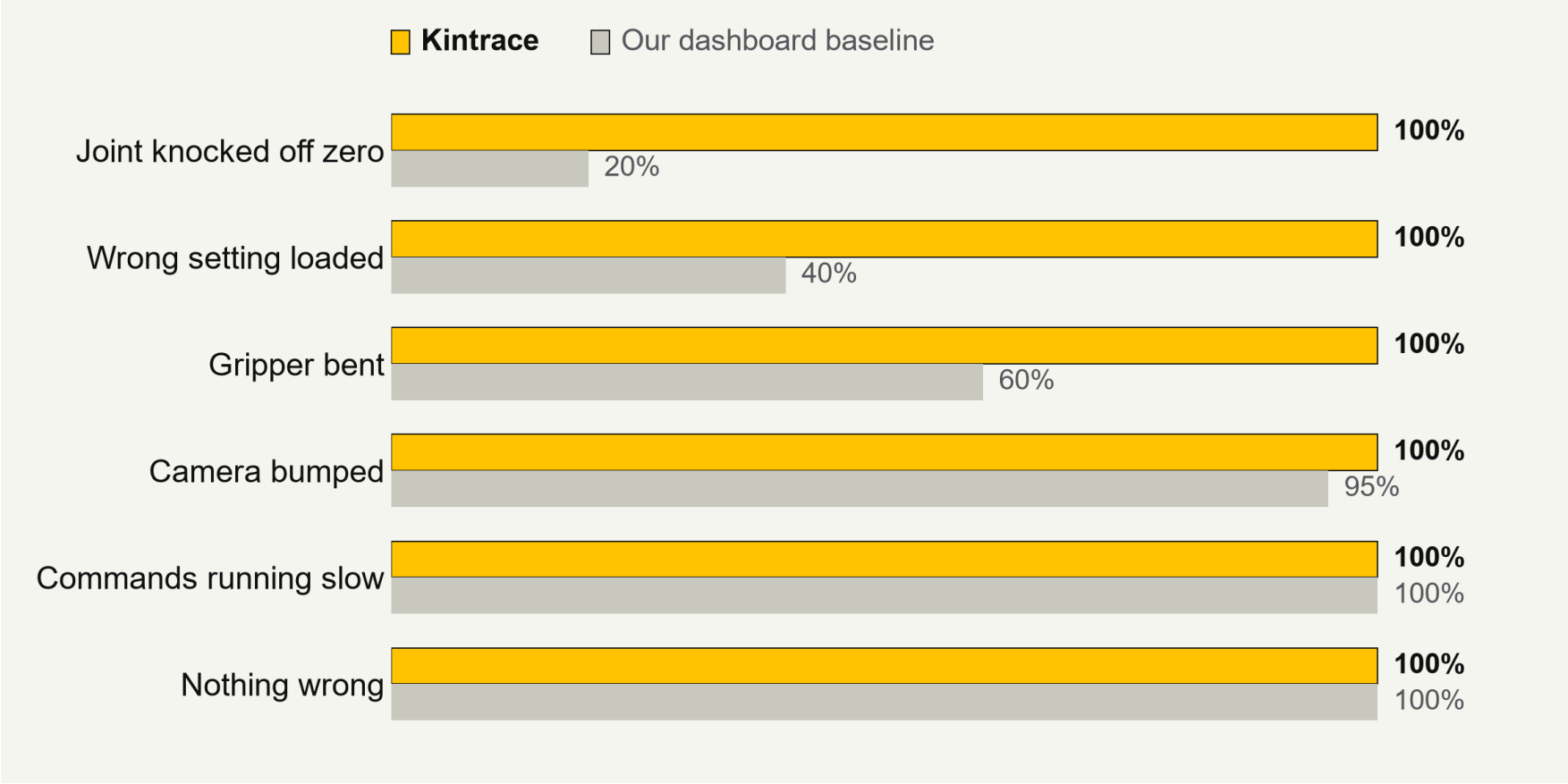
Explains

Names the cause and sizes it. Here: the camera, 18.1 mm, and nothing else moved.
- 3

Fixes

Writes the new calibration, re-checks, runs six test grabs, then signs GO or NO-GO.

In our simulator, Kintrace named the right cause every time. The dashboard baseline we wrote got 69%.



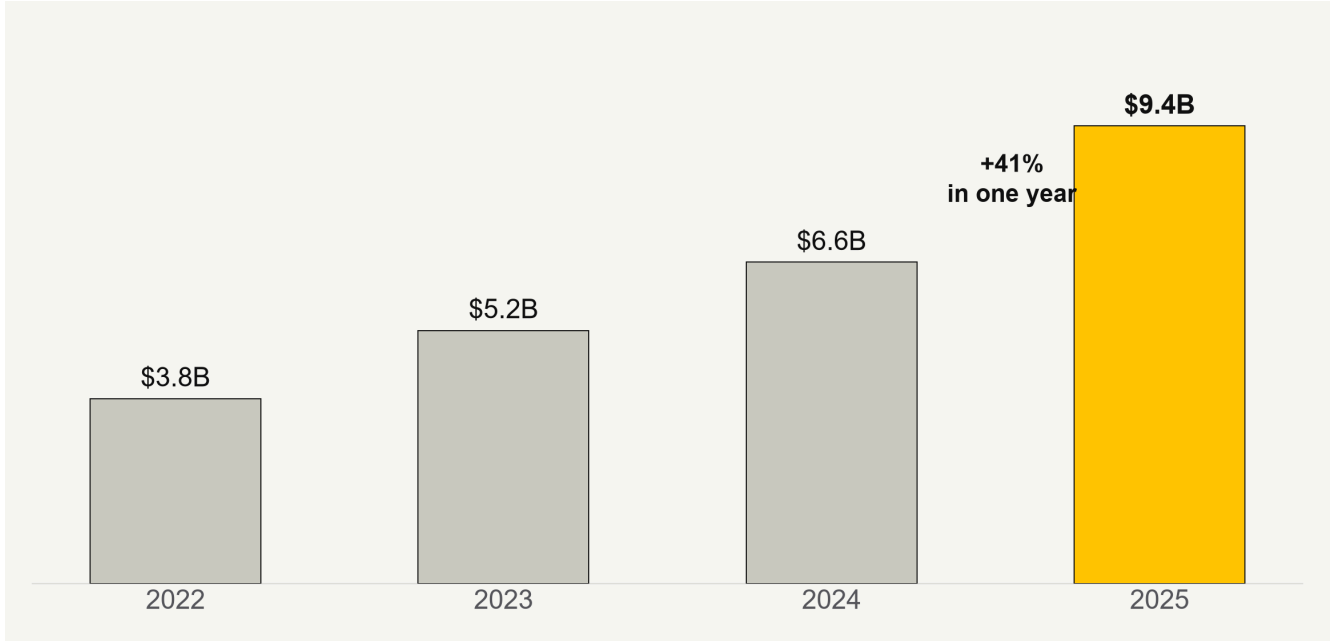
120 runs on our simulated UR5e pick cell, 20 per case, every fault sized to cause the same 20 mm miss. The baseline is ours, built from the signals a typical dashboard tracks. Never run on a real arm.

It also catches slow drift, about 55 seconds before the first grab misses.



One simulated run of a sagging camera mount. Across 20 drifts it warned before the first miss 19 times, named the cause right 20 times, and raised 0 false alarms in 230 healthy windows. Drift is sped up so a run takes minutes. We have not measured lead time at real-world drift rates.

There have never been more robots, and the money behind them doubled in three years.



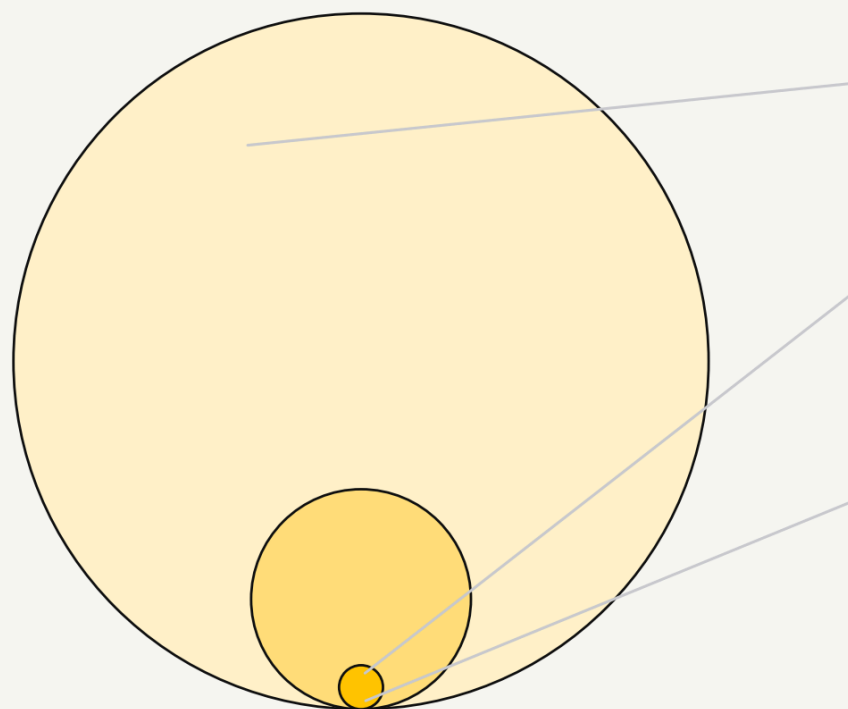
5 million

industrial robots now operating, a record, up 9% in a year. The US installed 38,500 and passed Japan. IFR, September 2026.

47% a year

forecast growth for physical AI, from \$0.89B in 2025 to \$15.2B by 2032. Every robot in it learns from recordings.

At \$200 a robot a month, the part we can realistically sell to is \$1.2B a year.



TAM \$12B a year

every industrial robot on earth
 $5,000,000 \times \$2,400$ a year

SAM \$1.2B a year

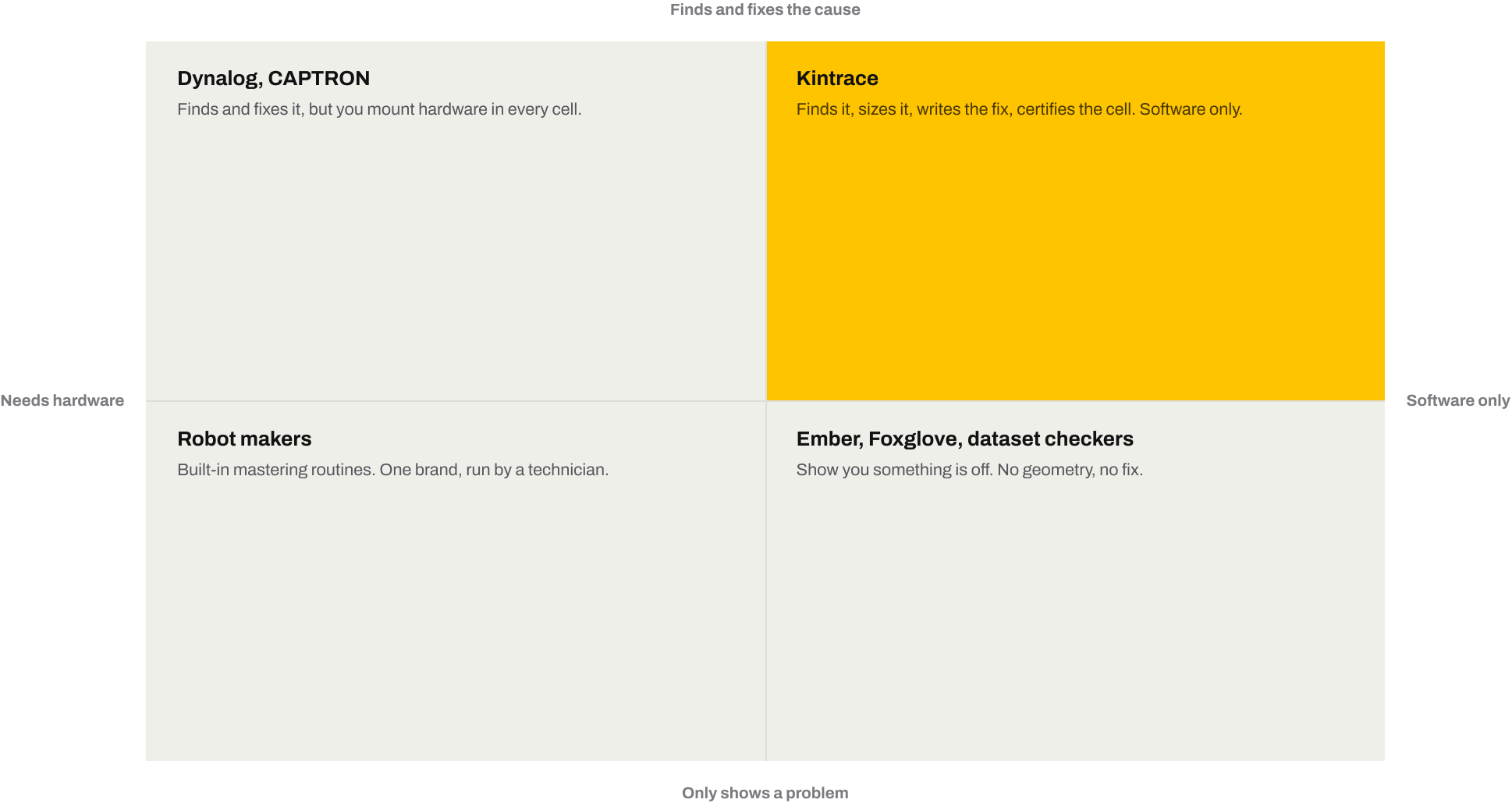
the 10% with a camera, in fleets
and labs we can reach (our estimate)

SOM \$48M a year

20,000 robots, five years out
 $20,000 \times \$2,400$ a year

Top down check: \$22.5B was spent on industrial robot services in 2023, heading to \$41.6B by 2033. Our \$12B bottom up number sits inside a services market already twice its size.

Software that names the cause and fixes it is an empty box.



Where we are going.



Now

The engine works in simulation. A live demo anyone can try, and the code is open on GitHub.



3 months

Kintrace running on a real robot arm, a video of it catching and fixing a bump, and first conversations with robot teams.



6 months

Three paid pilots with robot learning teams and robot fleets, and real robot data plugged in through ROS 2 logs.



12 months

Always-on monitoring for fleets, then working with robot makers to ship it on new arms.

Where this goes

Every robot arm checked the way a car has a check engine light, with Kintrace as the standard way to know a robot is physically right.

Why back Kintrace now.



It is built, not an idea

A working engine, a live demo, open code and 32 tests passing. You can run the whole thing yourself before you decide.



A big market with no software answer

5 million robots, 600,000 added a year, robot learning booming. Today you get a dashboard that only flags a problem, or hardware in every cell.



A hard technical edge

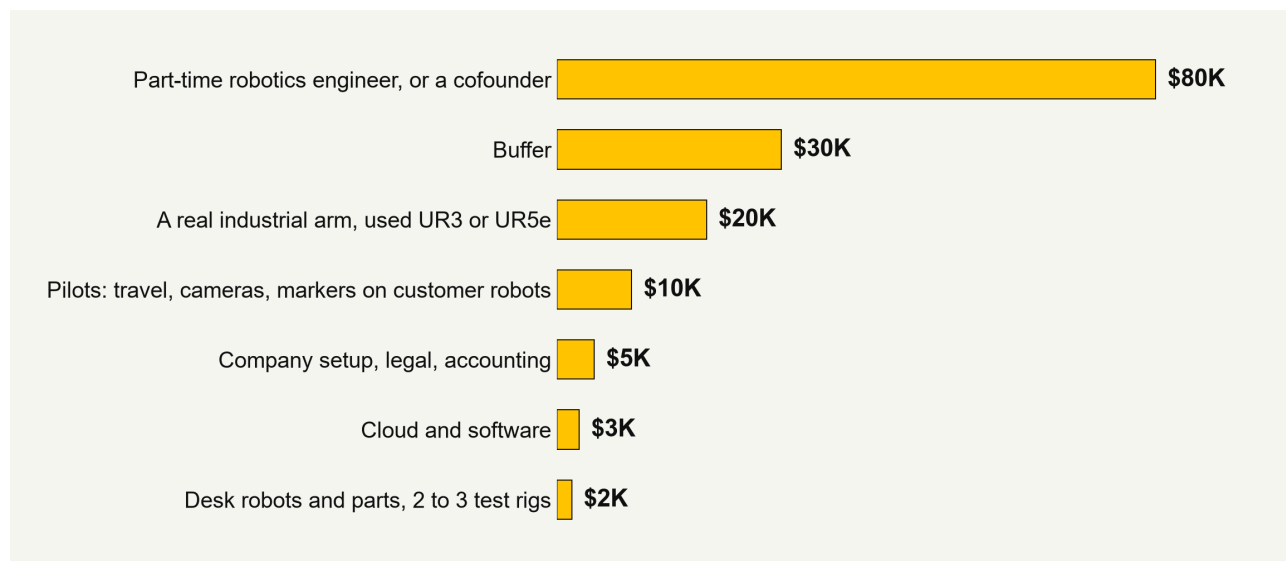
It tells apart faults that look identical from outside, camera against joint against tool, and gives the exact fix. 120/120 against 83/120 for a dashboard baseline. That is research, not a feature someone adds in a week.



The founder

Physical AI research at CMU, Harvard Innovation Labs. I built all of this myself, from idea to working product in weeks.

\$150K takes Kintrace from simulation to paying customers.



WHAT IT BUYS

- ☒ **3 months**
Kintrace on a real robot arm, a demo video, first user conversations
- ☐ **6 months**
Three paying pilots
- ☐ **12 months**
Results from real customers, ready for a seed round